

CSCE 645 GEOMETRIC MODELING · MODULE 07 · LECTURE NOTES

Discrete Curvature

Measuring bending on a surface that has none.

MODULE OUTCOMES

Define principal, mean, and Gaussian curvature for smooth surfaces.

Explain why naive discretisation fails on a mesh.

Compute discrete Gaussian curvature by angle defect.

Compute discrete mean curvature by the cotangent formula.

State the discrete Gauss–Bonnet theorem and use it as a test.

1 Curvature on a smooth surface

At a point p on a smooth surface with normal n , slice the surface with a plane containing n . The intersection is a curve, and it has a curvature. Rotating the plane around n gives a range of values; the maximum and minimum are the **principal curvatures** κ_1 and κ_2 , attained in orthogonal directions.

Quantity	Definition	Meaning	Sign
Mean curvature H	$(\kappa_1 + \kappa_2)/2$	Average bending; how far from a minimal surface.	Sign depends on the normal's orientation.
Gaussian curvature K	$\kappa_1 \cdot \kappa_2$	Intrinsic bending; how far from flat.	$K > 0$ dome or bowl; $K < 0$ saddle; $K = 0$ developable.

THEOREMA EGREGIUM: K IS INTRINSIC

Gauss's 'remarkable theorem': Gaussian curvature can be computed entirely from measurements made *within* the surface — lengths and angles — with no reference to the ambient space.

Roll a flat sheet of paper into a cylinder. Nothing stretched, so every intrinsic measurement is unchanged, and K remains 0 everywhere. But H changed, because the cylinder bends in space and the flat sheet did not. K is a property of the surface; H is a property of its embedding.

The consequence that matters for this course: a sphere has $K = 1/r^2 > 0$ and a plane has $K = 0$. Since K is intrinsic, no map from sphere to plane can preserve both lengths and angles — flattening *must* distort.

That is why every world map misrepresents something, and it is the reason UV unwrapping (Module 09) is a problem of choosing which distortion to accept rather than one of avoiding distortion.

2 Why a mesh has no curvature

Apply the smooth definitions pointwise to a triangle mesh:

- On the **interior of a face**, the surface is planar. Both principal curvatures are 0, so $H = K = 0$.
- Across an **edge**, the normal changes discontinuously. Curvature is a delta function — infinite, concentrated on a set of zero area.
- At a **vertex**, the normal is not even well defined.

So the answer is $\{0, \infty, \text{undefined}\}$ and carries no information about the shape. Discretising the formula does not work.

DISCRETISE THE THEOREM, NOT THE FORMULA

The method of discrete differential geometry is to choose a *structural property* of the smooth quantity — typically an integral identity or a conservation law — and define the discrete version so that the property holds **exactly** on a mesh, not approximately.

The resulting definition is then a genuine discrete analogue rather than a numerical approximation that happens to converge. Algorithms built on it inherit the structure, which is why they tend to be stable and to behave sensibly on coarse meshes.

There is no free lunch. No discrete definition can preserve every property of its smooth counterpart simultaneously — this is a theorem, not a limitation of current knowledge. Which property you preserve is a design decision, and different applications make different choices.

3 Discrete Gaussian curvature: angle defect

$$K_v = 2\pi - \sum_{\text{faces } f \text{ at } v} \theta_f$$

Sum the interior angles of all triangles meeting at a vertex and subtract from 2π . The intuition is immediate: if the angles sum to exactly 2π , the faces around the vertex can be flattened onto a plane without stretching, so the vertex is intrinsically flat. If they sum to less, there is a cone point — a dome. If more, there is excess material — a saddle.

$\Sigma\theta$	K_v	Shape
$= 2\pi$	0	Flat: the one-ring unfolds into a plane.
$< 2\pi$	> 0	Dome or cone point.
$> 2\pi$	< 0	Saddle.

WHY THIS DEFINITION IS THE RIGHT ONE

The smooth Gauss–Bonnet theorem states that for a closed surface, $\iint K \, dA = 2\pi\chi$ where $\chi = 2 - 2g$ is the Euler characteristic from Module 01.

With angle defect as the definition, the **discrete** Gauss–Bonnet theorem holds *exactly*:

$\sum_v K_v = 2\pi\chi$, for any mesh, at any resolution, with no error term whatsoever.

A four-triangle tetrahedron approximating a sphere gives exactly 4π . A million-triangle sphere gives exactly 4π . This is not convergence — it is an identity. That is what distinguishes a discrete *analogue* from an approximation.

USE IT AS YOUR TEST

Compute $\sum K_v$ on any closed mesh you load. For genus 0 it must equal $4\pi = 12.566\dots$ to within floating-point error. For a torus it must equal 0.

If it does not, something is wrong: the mesh has holes or non-manifold elements, the winding is inconsistent, or your angle computation is buggy. This single check has caught more bugs in this course than any other, and it is required in Project 1.

3.1 Integrated versus pointwise

Angle defect is an **integrated** quantity: it is the total curvature concentrated at the vertex, with units of a solid angle, not a curvature density. For a pointwise value comparable to the smooth K , divide by an associated vertex area:

$$K_{\text{pointwise}} = K_v / A_v$$

Confusing the two produces results that look right in shape and are wrong by a factor of the mesh resolution. Sanity-check on a sphere, where the pointwise value must be $1/r^2$ everywhere regardless of tessellation.

4 Discrete mean curvature: the cotangent formula

$$2 \mathbf{H}_v \mathbf{n}_v = \frac{1}{2} \sum_{j \in \mathcal{N}(v)} (\cot \alpha_{ij} + \cot \beta_{ij})(\mathbf{v}_i - \mathbf{v}_j)$$

where α_{ij} and β_{ij} are the two angles *opposite* the edge (i, j) in the two triangles sharing it.

THIS EXPRESSION IS THE DISCRETE LAPLACIAN

The right-hand side is the discrete Laplace–Beltrami operator applied to the position function — the subject of Module 08, appearing here first.

The smooth identity it discretises is $\Delta \mathbf{x} = 2\mathbf{H}\mathbf{n}$: the Laplacian of the position function equals mean curvature times the normal. Mean curvature and the Laplacian are not two things that happen to be related; they are the same object.

The practical consequence is immediate and is the reason Module 08 follows directly: **smoothing a surface and reducing its mean curvature are the same operation**. Laplacian smoothing is literally mean curvature flow.

4.1 Where the cotangents come from, and where they bite

The weights are not chosen for convenience. They arise from a finite element discretisation of the Laplacian using piecewise-linear hat functions on the triangulation, and they make the discrete operator agree with the smooth one in a precise variational sense — it is the gradient of the discrete Dirichlet energy.

OBTUSE TRIANGLES PRODUCE NEGATIVE WEIGHTS

The cotangent of an obtuse angle is negative. An edge opposite two obtuse angles therefore receives a **negative weight**.

A negative weight means the neighbour pushes the vertex *away* rather than pulling it, so Laplacian smoothing can move a vertex outward, the operator loses the maximum principle, and the resulting linear systems can lose positive-definiteness.

Symptoms are distinctive: smoothing that increases noise in some region, parameterisations with flipped triangles (Module 09), or solvers that fail to converge.

Mesh quality is not cosmetic. It determines whether your operators behave at all, which is the strongest argument for the remeshing of Module 10. The alternatives — clamping the weights to zero, or using the intrinsic Delaunay triangulation — both trade accuracy for robustness.

5 Computing it reliably

Decision	Options	Recommendation
Vertex area	Barycentric (one third of each incident triangle); Voronoi; mixed Voronoi-barycentric.	Mixed. It handles obtuse triangles correctly, where pure Voronoi areas can fall outside the triangle.
Integrated or pointwise	Angle defect is integrated by nature.	Keep both; be explicit in your API about which you are returning. Most bugs here are unit confusion.
Validation	Sphere: $K = 1/r^2$, $H = 1/r$. Torus: $\Sigma K = 0$. Plane: both zero.	Test on all three before trusting the code on real data.
Gauss–Bonnet check	$\Sigma K_v = 2\pi\chi$.	Assert it. Non-negotiable.
Obtuse triangles	Leave, clamp, or remesh.	Detect and report them. Remesh if the downstream algorithm is sensitive.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Keenan Crane — DDG, curvature lectures and notes	The primary source for this module. The 'discretise the theorem' philosophy is developed properly, with the angle defect derivation.
Meyer, Desbrun, Schröder & Barr — Discrete Differential-Geometry Operators (free)	The paper that established the cotangent formula and the mixed area region in graphics. Short and still the standard reference.
Wardetzky et al. — Discrete Laplace operators: no free lunch (free)	The proof that no discrete Laplacian preserves all the smooth properties. Three pages, and it reframes the whole subject.
libigl tutorial — curvature	Working implementations to check your own against.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement discrete Gaussian curvature by angle defect. Verify that $\Sigma K_v = 4\pi$ on a sphere mesh and 0 on a torus. Report both numbers to six decimal places.
- 2 Verify that Gauss–Bonnet holds at every resolution: compute the sum on sphere meshes of 20, 200, 2,000, and 200,000 triangles and confirm all four give 4π .

- 3 Compute pointwise Gaussian curvature on a sphere of radius r and confirm it equals $1/r^2$ everywhere. Identify and explain any deviation at irregular vertices.
- 4 Implement the cotangent formula for mean curvature. Validate on a sphere ($H = 1/r$) and a cylinder ($H = 1/2r$).
- 5 Visualise both curvatures as per-vertex colour on a scanned model. Identify ridges, valleys, and saddles by eye and confirm against the sign of K .
- 6 Detect obtuse triangles in a mesh and count how many produce negative cotangent weights. Then apply Laplacian smoothing and look for vertices that move the wrong way.
- 7 Implement all three vertex-area definitions and compare the pointwise curvature each produces on a mesh with some obtuse triangles.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define principal, mean, and Gaussian curvature.
- 2 State Theorema Egregium and explain why it means UV unwrapping must distort.
- 3 Why does applying the smooth curvature definitions pointwise to a mesh give no information?
- 4 Explain 'discretise the theorem, not the formula', and state what the no-free-lunch result says.
- 5 Give the angle defect formula and explain why it is the correct discrete Gaussian curvature.
- 6 State discrete Gauss–Bonnet and say why it is the best test in this course.
- 7 What identity connects mean curvature and the Laplacian, and what practical consequence follows?
- 8 Why do obtuse triangles cause trouble, and what are your options?